

Quiz (Carolina Reaper)

Q1. What is the greatest common factor of 12 and 18?

- (A) 2
- (B) 3
- (C) 6
- (D) 9
- (E) 12

Q2. Factor out the greatest common factor from $6x + 12y$

- (A) $2(3x + 6y)$
- (B) $3(2x + 4y)$
- (C) $6(x + 2y)$
- (D) $12(x + y)$
- (E) $6(x + 2y)$

Q3. What is the greatest common factor of 24 and 30?

- (A) 2
- (B) 3
- (C) 6
- (D) 12
- (E) 18

Q4. Simplify the expression $9x^2 + 12x$ by factoring

- (A) $3x(3x + 4)$
- (B) $3x(3x + 2)$
- (C) $3x^2(3 + 4)$
- (D) $3(3x^2 + 4x)$
- (E) $9x(x + 2)$

Q5. What is the greatest common factor of 15 and 25?

- (A) 1
- (B) 3
- (C) 5
- (D) 10
- (E) 15

Q6. Factor out the greatest common factor from $8x^2 + 16x + 24$

- (A) $8(x^2 + 2x + 3)$
- (B) $4(2x^2 + 4x + 6)$
- (C) $8(x^2 + 2x + 3)$
- (D) $2(4x^2 + 8x + 12)$
- (E) $4(2x^2 + 4x + 4)$

Q7. Simplify the expression $12x^2y + 15xy$ by factoring

- (A) $3xy(4x + 5)$
- (B) $3x(4xy + 5y)$
- (C) $3xy(4x + 3)$
- (D) $3(4x^2y + 5xy)$
- (E) $3xy(4x + 4)$

Q8. What is the greatest common factor of 20 and 35?

- (A) 1
- (B) 2
- (C) 5
- (D) 7
- (E) 10

Open Ended Questions (FRQ)

Q9. Factor out the greatest common factor from $10x^3 + 20x^2 + 30x$, show all steps and simplify

Q10. Simplify the expression $18x^2 + 12x + 6$ by factoring, show all steps

Chef's Tips

- Tip 1: Use the distributive property in reverse to factor out the GCF
- Tip 2: Write out all the factors of each term to find the GCF
- Tip 3: Check your answers by distributing the GCF back to the terms